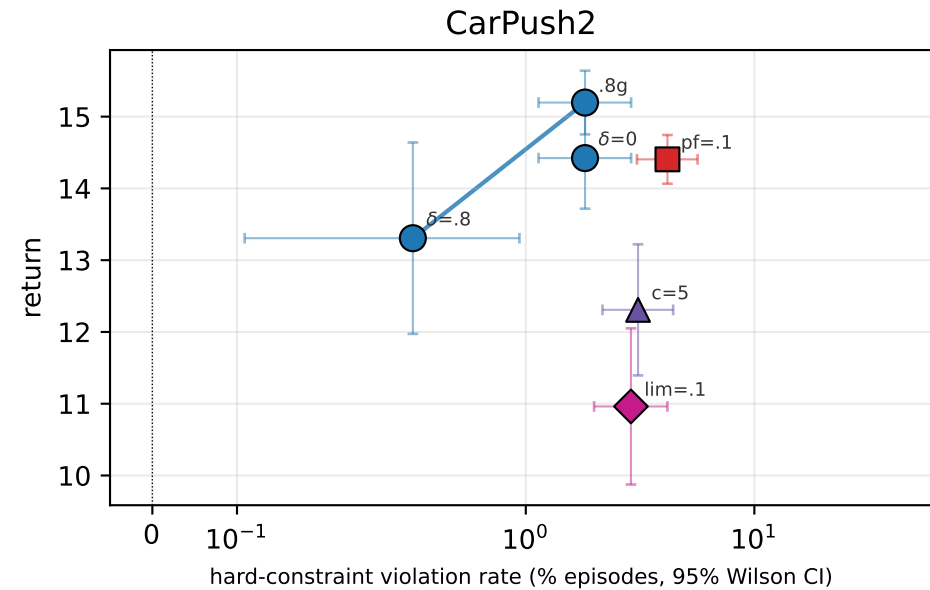
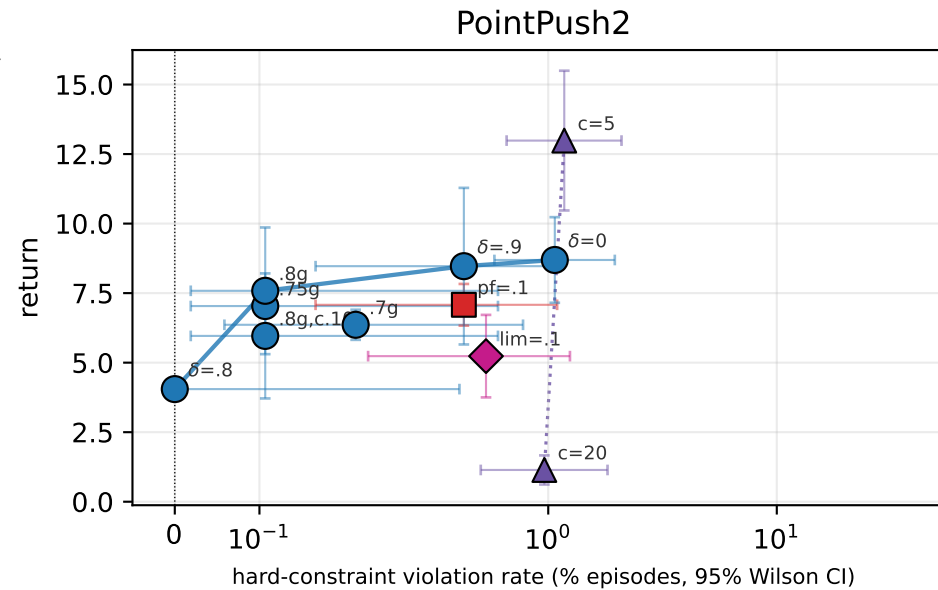
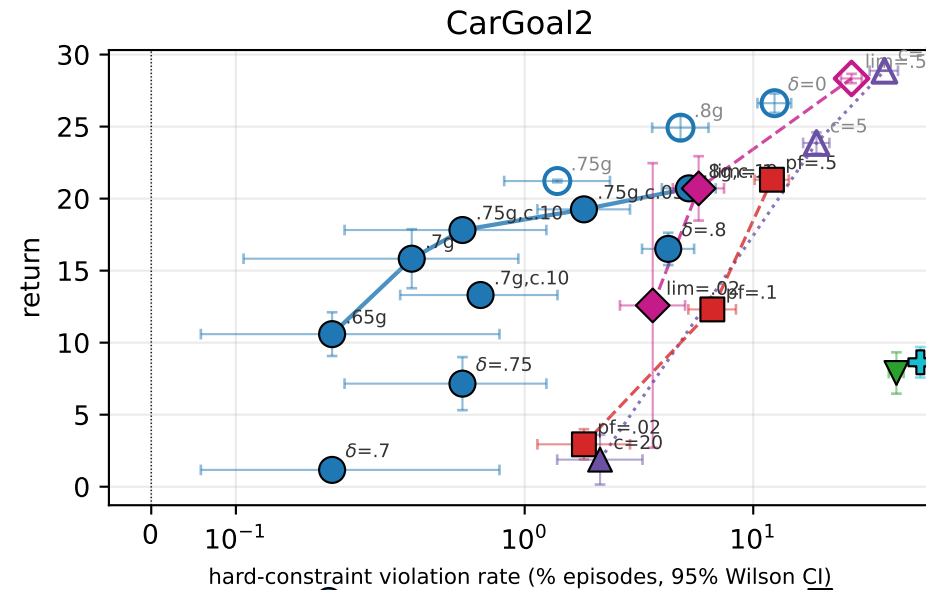


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



● SafeQT-typed (ours)

SafeOT-typed (ours)

- Safety typed (Gals)
- FDBL (constraint based)

FDPI (constraint-based)

◆ SAC-Lagrangian (penalty-based)

SDAC (multi-constraint SOTA)

 SDAC (multi-constraint SOTA)
 SBCBO (multi-constraint SOTA)

- SRCPO (multi-constraint SOTA)

— typed Pareto frontier (soft ≤ 25)

TA) SAC-Pen coefficient sweep

5DPI n. sweep

--- FDPI p_f sweep

- SAC-Lag cost-limit sweep